

Ordinary Differential Equations

Mid-Semester Exam

Total Points: 30

Time: 90 minutes

Important

1. Simplify your answers as much as possible and write them in terms of fractions or constants such as $\sqrt{e}$ or $\log(4)$ rather than decimals.
 2. Show all your work and explain your reasonings clearly.
 3. No books, notes, calculators or any papers are allowed. Additional paper is available upon request.
 4. You are not allowed to communicate with other students during the exam. Copying will not be tolerated.
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1. (6 points) Find the *general solution* of the following ODEs.

(a) $(e^{2t} - t^2 \sin x) \frac{dx}{dt} + 2xe^{2t} + 2t \cos x = 0.$ (3 points)

(b) $\frac{dx}{dt} = x \log\left(\frac{1}{x}\right).$ (3 points)

2. (6 points) Solve the following *initial value problems*.

(a) $\frac{dx}{dt} = \frac{t}{1+t}x + 1$, with $x(0) = 0.$ (3 points)

(b) $\frac{dx}{dt} = (x - \sin t)^8 + \cos t$, with $x(0) = 0.$ (3 points)

3. (3 points) Evaluate $\lim_{t \rightarrow +\infty} x(t)$, where $x(t)$ satisfies: $\dot{x}(t) = (\cos t - 2)x(t) + (1 + t^6)^{-1}.$

4. (5 points) Construct *three distinct solutions* of the the initial value problem: $\begin{cases} \dot{x} = |x|^{3/4} \\ x(0) = 0. \end{cases}$

5. (3 points) Show that the initial value problem: $\begin{cases} \dot{x} = \frac{3e^{-t^2}}{1+x^4} + \frac{x \sin x}{2+t^2} \\ x(0) = 0. \end{cases}$ has a *unique solution defined for all* $t \in \mathbb{R}.$

6. (4 points) Suppose $o(t)$ satisfies the initial value problem: $\begin{cases} \dot{x} = (1 - x^2) \sin(1 + x^2) \\ x(0) = 0. \end{cases}$

Show that $o(t)$ is *defined for all* $t \in \mathbb{R}$ and $|o(t)| \leq 1$ for all $t \in \mathbb{R}.$

7. (3 points) Suppose $x(t)$ satisfies the differential equation: $2\dot{x} = (2 \cos(2x) + 3)x.$

Show that

$$x^2(t) \leq x^2(0) e^{5t} \text{ for all } t \geq 0.$$

Ordinary Differential Equations

End-Semester Exam

Total Points: 50
Time: 150 minutes

Important

1. Simplify your answers as much as possible and write them in terms of fractions or constants such as $\sqrt{e}$ or $\log(4)$ rather than decimals.
2. Show all your work and explain your reasonings clearly.
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1. (9 points)

- (a) Find the general solution $y(x)$ of: $xy' = y + \sqrt{x^2 + y^2}$.
- (b) Suppose $\phi(t)$ satisfies: $\phi'(t) = \cos \phi(t)$, with $\phi(0) = 0$. Show that:

$$\log(1 + \sin \phi(t)) = \log(1 - \sin \phi(t)) + 2t.$$

- (c) Find the general solution $x(t)$ of: $x'' + 2x' + 2x = e^t$.

2. (5 points) Let $A, B > 0$ and consider:

$$x' = Bx - Ax^2.$$

- (a) Find three distinct solutions of the above equation.
- (b) For every solution $x(t)$ of the above equation with $x(0) > 0$ show that $x(t) > 0$ for all $t \geq 0$ and $\lim_{t \rightarrow +\infty} x(t) = \frac{B}{A}$.

3. (3 points)

- (a) Determine if there exists a unique function $x(t)$ satisfying: $x' = \begin{cases} 1 & \text{if } x < 0 \\ -1 & \text{if } x \geq 0 \end{cases}$ for $t > 0$, with $x(0) = 1$.
- (b) Determine if there exists a unique function $f(t)$ satisfying: $(1+t)f'(t) + \sin(t + e^{-f(t)}) = 0$ for $t > 0$, with $f(0) = 1$.

4. (4 points) Let $Q(t) = \sum_{k=0}^n (2t^3 + 3t^2)^k$, and consider the initial value problem:

$$x'(t) = Q(\cos(t)), \quad \text{with } x(0) = x_0 \in \mathbb{R}.$$

- (a) Show that the above initial value problem has a unique solution for any initial data $x_0 \in \mathbb{R}$.

- (b) Determine the maximal interval of existence for the above initial value problem with initial data $x_0 \in \mathbb{R}$.

5. (8 points)

- (a) Solve the initial value problem: $\begin{cases} x' = -y - t & \text{with } x(0) = 1 \\ y' = -x + t & \text{with } y(0) = 0. \end{cases}$

- (b) Find the general solution of the system: $X'(t) = \begin{pmatrix} a & b \\ c & a \end{pmatrix} X(t)$, where $bc > 0$.

6. (4 points) Consider the system: $X'(t) = \begin{pmatrix} \alpha & \beta \\ \beta & 2 \end{pmatrix} X(t)$, where $\alpha, \beta \in \mathbb{R}$. Find the condition on the parameters α, β so that:

- (a) Origin is the only equilibrium point for the above linear system
 (b) And then, origin is a source for the above linear system.

7. (5 points) Let P and Q be continuous functions on $[a, b]$ and suppose ϕ_1 and ϕ_2 satisfies:

$$x''(t) + P(t)x'(t) + Q(t)x(t) = 0 \quad \text{for } t \in (a, b).$$

- (a) If ϕ_1 attains its minimum and ϕ_2 attains its maximum at some $\xi_0 \in (a, b)$, then show that ϕ_1 and ϕ_2 cannot form a fundamental set of solutions.
 (b) Suppose ϕ_1 and ϕ_2 are two linearly independent solutions for the above equation. Show that the Wronskian

$$W(\phi_1, \phi_2)(t) := \begin{vmatrix} \phi_1(t) & \phi_2(t) \\ \phi_1'(t) & \phi_2'(t) \end{vmatrix}$$

satisfies the equation

$$W'(t) + p(t)W(t) = 0 \quad \text{for } t \in (a, b).$$

8. (5 points) Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function with $g(0) = 0$ and $xg(x) > 0$ for all $x \neq 0$. Consider the second order ODE:

$$x''(t) + g(x(t)) = 0 \quad \text{for } |t| < 2.$$

- (a) Find the equilibrium solutions of the above equation.
 (b) Determine the stability behaviour of the equilibrium solutions of the above equation.

9. (7 points) Determine all equilibrium solutions and their stability behaviour for the following nonlinear systems:

(a) $\begin{cases} x'(t) = 2 \sin(x(t)) - 4y(t) \\ y'(t) = \sin(x(t)) - 3y(t) \end{cases} \quad \text{for all } t \in \mathbb{R}.$

(b) $\begin{cases} x'(t) = x^2(t) + y^2(t) - 1 \\ y'(t) = -2y(t). \end{cases} \quad \text{for all } t \in \mathbb{R}.$